Redesigning Florence Nightingale's Coxcomb Diagram Using Contemporary Interactive Methods

Introduction

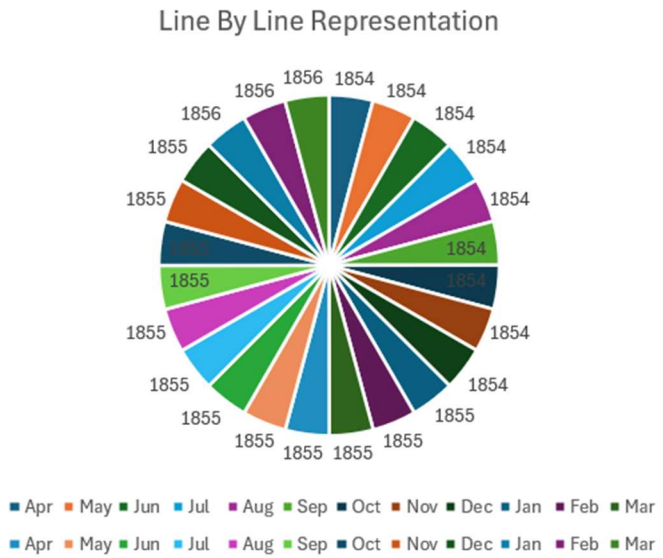
Florence Nightingale's coxcomb diagram stands as one of the most iconic examples of data visualization used for social reform. Its radial form and bold color segmentation were persuasive tools in the 19th century, effectively communicating the disproportionate number of deaths caused by preventable diseases.

However, contemporary audiences often benefit from more structured and interactive representations. Modern visualization tools allow us to explore temporal patterns, compare values precisely, and engage with data dynamically. This project reinterprets Nightingale's work through a series of design iterations, culminating in an interactive stacked bar chart that preserves the original message while enhancing clarity and usability.

Iteration 1 — Full Dataset Display

The first iteration attempted to visualize the entire two-year dataset in a single static view. Although this approach captured all the data, it resulted in a dense and visually overwhelming graphic. Without clear grouping or temporal segmentation, the audience would struggle to extract meaningful insights.

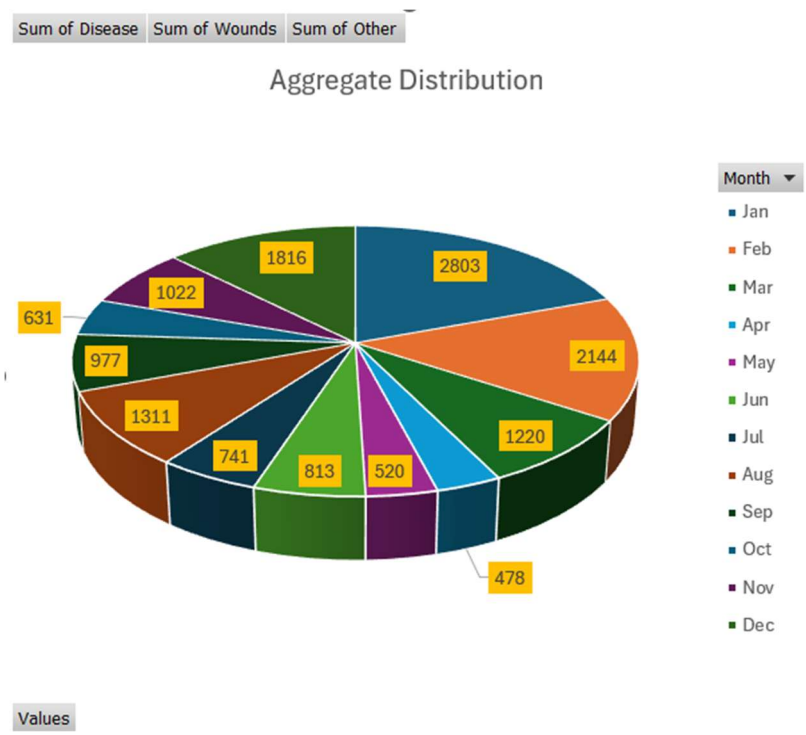
A visualization must balance completeness with readability. Simply displaying all data points does not guarantee effective communication.



Iteration 2 — Monthly Grouping Using a Pie Chart

The second iteration grouped the data by month excluding the year and represented the totals using a pie chart. This produced a clean and visually appealing summary of the proportions across the three categories—Disease, Wounds, and Other Causes.

Because pie charts show proportions at a single point in time, they cannot convey how mortality patterns changed across the two-year period. This limitation made the pie chart unsuitable for communicating the month-to-month variation that was central to Nightingale's original argument.



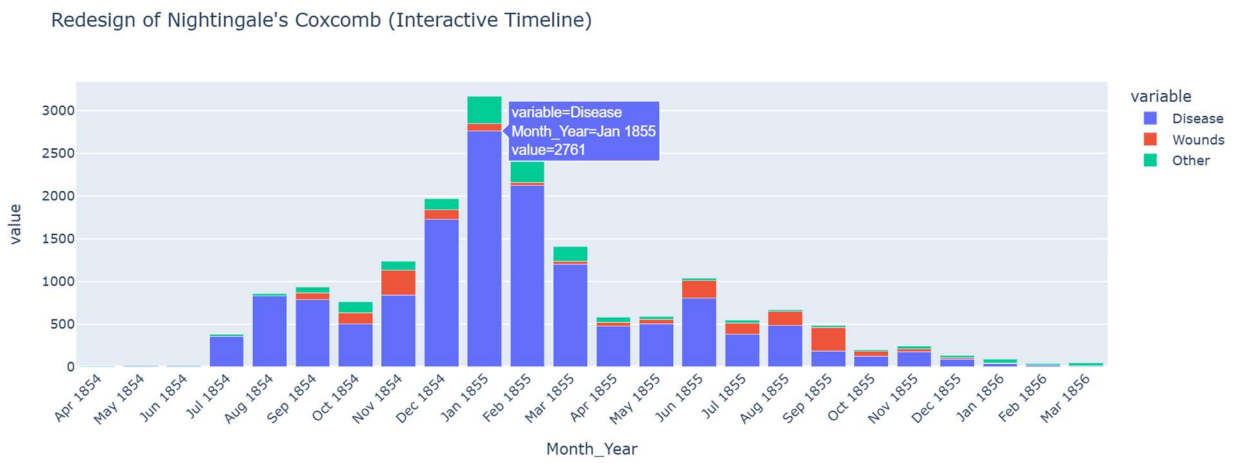
Iteration 3 — Final Design Using an Interactive Bar Chart (Plotly)

The final iteration uses an interactive stacked bar chart created with Python and Plotly. This approach preserves the temporal structure of the dataset while making the information easier to interpret.

By combining the month and year into a single Month_Year label, the visualization presents a clear 24-month timeline from April 1854 to March 1856. The stacked bars maintain the

categorical breakdown (Disease, Wounds, Other Causes), while interactive features—such as hover tooltips and zooming—allow users to explore the data in detail.

This design successfully communicates the underlying message of Nightingale's original chart while leveraging modern visualization techniques to enhance clarity, engagement, and accessibility.



Conclusion

This redesign demonstrates how historical visualizations can be reinterpreted using modern tools and interaction techniques. While Nightingale's coxcomb chart was persuasive and innovative for its time, contemporary audiences benefit from clearer, more structured, and interactive representations.

Through iterative exploration—moving from a full dataset display, to a pie chart, and finally to an interactive stacked bar chart—the project highlights how design decisions shape understanding. The final visualization preserves the spirit of Nightingale's message while making the data more accessible and engaging for today's viewers.

```
# Import required libraries
import pandas as pd
import plotly.express as px
import matplotlib.pyplot as plt

# Load the historical Nightingale dataset from CSV
df = pd.read_csv("nightingale_data.csv")

# Display the first few rows to confirm the data loaded correctly
df.head()

# Create a combined Month-Year label so the timeline is explicit and prevents months from different years from
overlapping in the chart
df["Month_Year"] = df["Month"] + " " + df["Year"].astype(str)

# Build an interactive stacked bar chart using Plotly
# - x-axis: Month-Year timeline (24 months from Apr 1854 to Mar 1856)
# - y-axis: Three causes of death (stacked)
# - barmode="stack": visually similar to Nightingale’s categorical grouping
fig = px.bar(
   df,
   x="Month_Year",
   y=["Disease", "Wounds", "Other"],
   barmode="stack",
   title="Redesign of Nightingale's Coxcomb (Interactive Timeline)"
)

# Rotate x-axis labels for readability since there are 24 entries
fig.update_layout(xaxis_tickangle=-45)

# Display the interactive visualization
fig.show()
```